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IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID: 25-26J-374

1. Topic (12 words max)

An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Teams are usually manually assigned or self-selected based on friendships, without considering interests, technical proficiency, academic prowess, or compatibility in working methods. As a result, there are internal disagreements, uneven workloads, poor teamwork, and decreased motivation in poorly balanced teams. The lack of ability of students to assess and create appropriate study ideas is another prevalent but sometimes overlooked problem.

Many students submit plans for projects that are unreliable, technically impossible, or inappropriate without early expert help or a structured topic evaluation, wasting time and requiring repeated changes. Many students experience issues with academic mentoring during the project, in addition to issues pertaining to teams and topics. Insufficient feedback and limited domain-specific support results from the fact that supervisors are frequently assigned based on availability rather than project-topic alignment. Students usually lack prompt direction, particularly when they are not in regular sessions, which hinders their progress and ability to make decisions. Furthermore, real-time tools to track individual contributions, team performance, and early warning indicators like missed deadlines, low productivity, or interpersonal problems are frequently lacking at institutions. Without prompt academic help, these unidentified problems may worsen, leading to unsuccessful or delayed projects and lowered learning objectives.

References

I. Wu, C., & Hou, H. (2021). Group formation using machine learning for educational team projects. IEEE Transactions on Learning Technologies, 14(3), 301 –313.

II. Devlin, J., Chang, M. -W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. NAACL -HLT.

III. Huo, Y., & Zhang, X. (2023). Enhancing academic supervision through AI-based mentor matching. International Journal of Educational Technology, 9(2), 89 – 97.

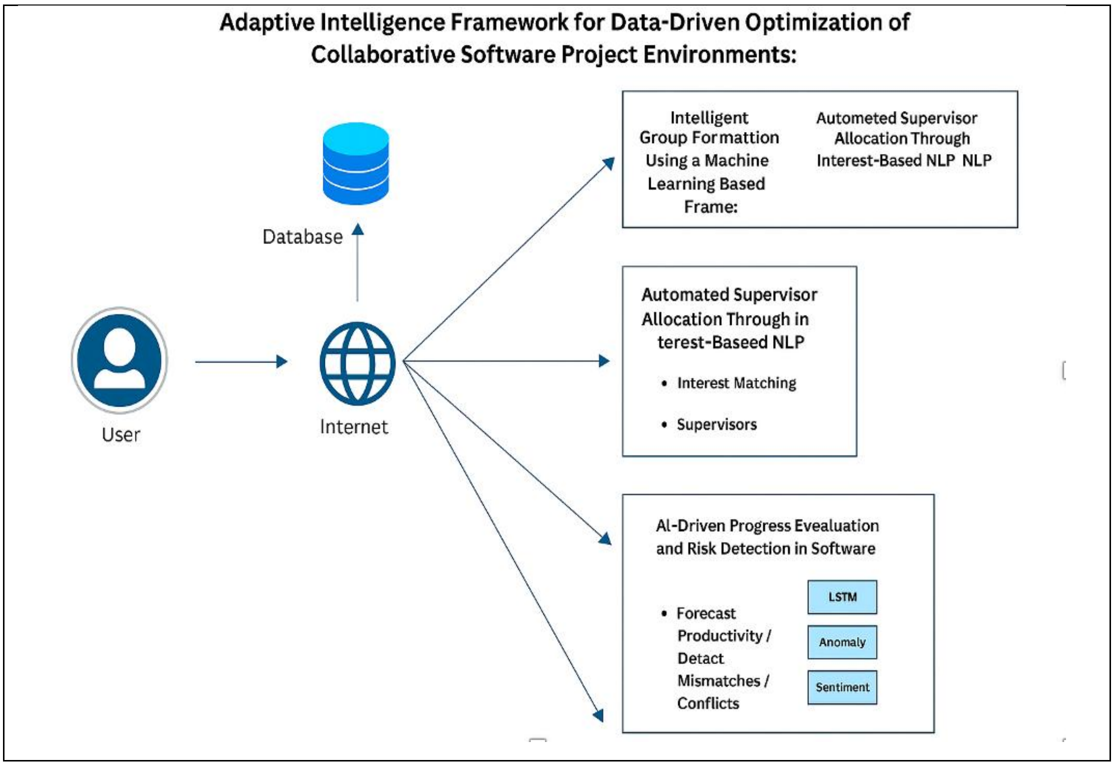
IV. Lipton, Z. C., Kale, D. C., & Wetzel, R. (2016). Learning to diagnose with LSTM recurrent neural networks. arXiv preprint arXiv:1511.03677.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is an end-to-end AI-powered academic project management system that addresses three core challenges: optimal group formation, supervisor-student topic matching, and continuous progress evaluation.

The system operates in three integrated modules:

1. Intelligent Group Formation: Student data, including academic records, skill assessments, and personality traits (via psychometric analysis), are collected. Unsupervised learning techniques such as K-Means or Hierarchical Clustering are applied to form balanced teams. Simultaneously, LLMs (like GPT) evaluate and refine submitted project topics to ensure technical and practical relevance.
2. Supervisor-Student Matching System: Using NLP techniques, the system semantically analyzes both student project descriptions and supervisor research profiles. A matching algorithm ranks supervisors based on relevance, availability, and workload. A chatbot powered by neural networks supports students with real-time guidance, resource links, and project advice.
3. AI-Driven Progress Monitoring: Throughout the project timeline, team progress and communication logs (e.g., status reports, messages) are analyzed using LSTM networks and sentiment analysis. This enables early detection of risks such as poor collaboration or low productivity. Alerts and recommendations are generated to both students and supervisor.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

An Adaptive Intelligence Framework for Data-Driven Optimization of Collaborative Software Project Environments—requires specialized expertise in various interrelated areas.

Machine Learning & AI: The framework heavily relies on clustering (for group formation), natural language processing (for supervisor matching and mentorship communication), and time series forecasting (for risk detection). Familiarity with models like K-Means, LDA, LSTM, and transformer-based language models is essential.

Natural Language Processing (NLP): NLP plays a central role in automating supervisor allocation, analyzing mentor-student communication, and summarizing text. Expertise in semantic similarity analysis, entity recognition, and dialogue modeling is critical for effective implementation.

Team Behavior and Sentiment Analysis: The progress monitoring component involves sentiment tracking and anomaly detection. This requires an understanding of behavioral analytics, team dynamics, and techniques like semantic sentiment extraction and productivity pattern recognition.

Educational Data Modeling: Understanding academic structures, student performance metrics, and project-based learning is important to ensure group formation and supervisory alignment are academically meaningful.

Data Requirements:

• Student Profiles: Academic records, skill assessments, and psychometric attributes for clustering.

• Supervisor Metadata: Fields of expertise, availability, historical performance, and research alignment.

• Project Records: Structured information on project descriptions and outcomes to facilitate matching.

• Chat Data: Annotated mentor-student interactions for training NLP and recommendation models.

• Team Performance Logs: Time-stamped task completions, communication patterns, and milestone tracking for LSTM modeling and risk analysis.

8. Objectives and Novelty

Main Objective			
To design and develop an intelligent AI-driven academic project management system that automates group formation, matches students with suitable supervisors based on project topics and expertise, and continuously evaluates team performance using advanced machine learning, NLP, and deep learning techniques to enhance academic efficiency, collaboration, and outcomes.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Komila K IT22029218	Intelligent Group Formation Using Machine Learning and Topic Evaluation Using LLM	- Apply clustering techniques to group students based on academic profiles, skills, and traits. - Use LLMs to evaluate and fine-tune project topics.	Automatic optimal group formation through multi-dimensional data analysis combined with LLM-driven topic refinement.
Mayoorabi U IT22322494	Supervisor-Student Matching and NLP-Based Project Guidance System	- Implement NLP models to match student topics with supervisor expertise. - Develop a neural network-based chatbot for real-time project guidance.	Interest-based supervisor allocation and dynamic chatbot support for personalized project mentoring.

Mohanakumar N IT22919564	AI-Driven Progress Evaluation and Risk Detection	- Use LSTM models to forecast team performance and identify risks. - Apply sentiment analysis and anomaly detection to flag productivity gaps.	Predictive monitoring and proactive intervention using real-time analytics for academic project management.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22029218	The component "Intelligent Group Formation Using Machine Learning and Topic Evaluation Using LLM" aligns with the specialization in Machine Learning and Natural Language Processing. It involves applying unsupervised learning algorithms such as K-Means and Hierarchical Clustering to analyze and group students based on academic performance, technical skills, and personality traits. The integration of Large Language Models (LLMs) for evaluating and refining project topics demonstrates expertise in NLP and generative AI, directly applying the core concepts of AI model training, fine-tuning, and prompt engineering.
Mayoorabi U IT22322494	The component "Supervisor-Student Matching and NLP-Based Project Guidance System" complies with the specialization in Artificial Intelligence and Deep Learning. It involves developing an NLP-based semantic matching engine that matches student-submitted project topics with supervisor profiles. The use of advanced language models and the development of a chatbot based on neural networks demonstrate practical knowledge of deep learning architectures (e.g., RNNs, Transformers) and conversational AI, aligning well with AI system design and deployment in educational technology.
Mohanakumar N IT22919564	The component "AI-Driven Progress Evaluation and Risk Detection" is closely aligned with the specialization in Deep Learning and Predictive Analytics. This component uses LSTM (Long Short-Term Memory) models for time-series forecasting of team performance and sentiment analysis for interpreting group communication. The implementation of anomaly detection methods for identifying risks early reflects a strong grasp of deep learning in real-time systems, predictive modeling, and proactive analytics, which are key competencies in AI-driven monitoring systems.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Prof	Samantha	Rajap[illegible]	[illegible]
Co-Supervisor	Mrs	Thamalee	kelegama	[illegible]
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

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Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.